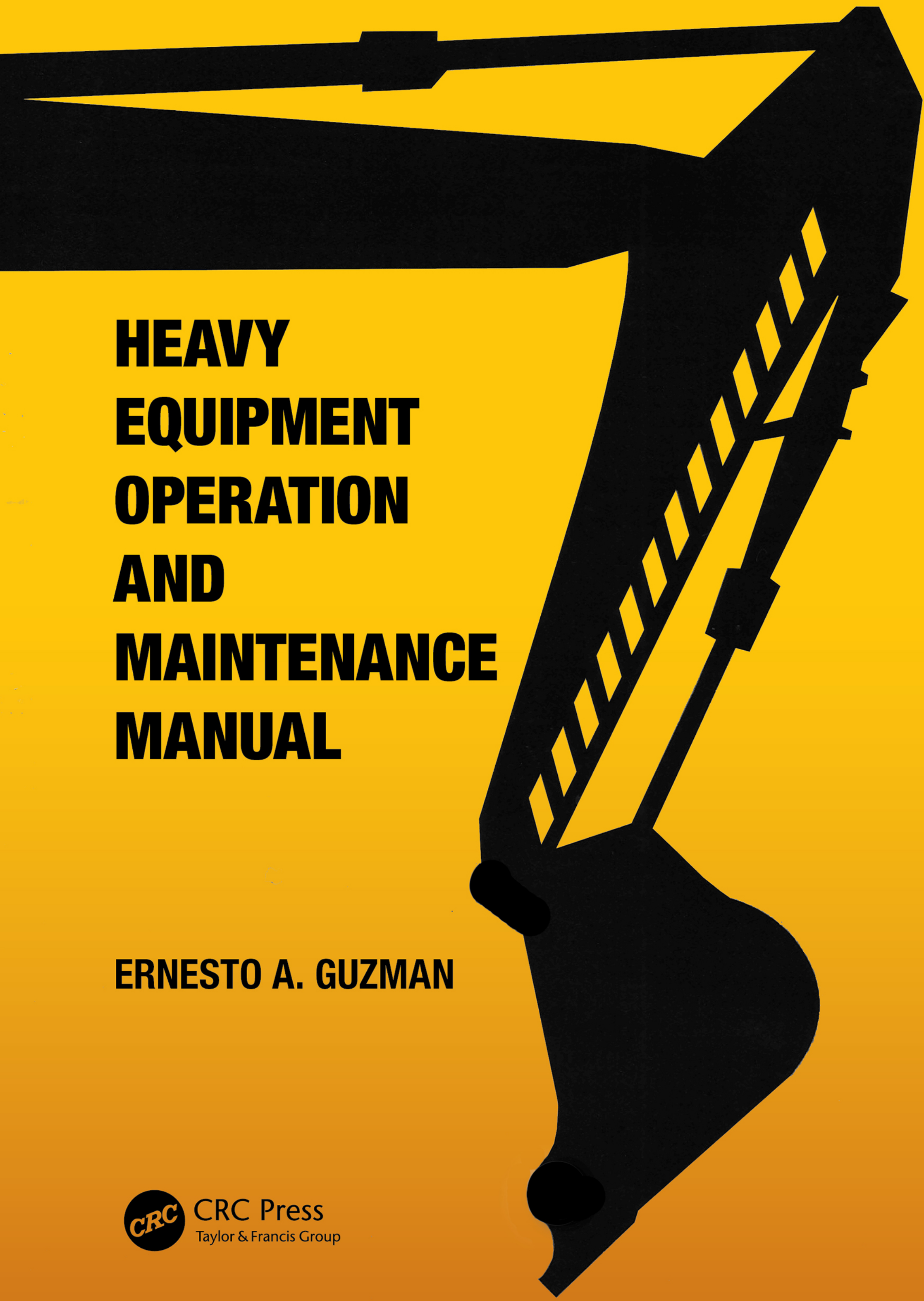
HEAVY EQUIPMENT OPERATION AND MAINTENANCE MANUAL



ERNESTO A. GUZMAN

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Heavy Equipment Operation and Maintenance Manual

Starting from the purchase of heavy equipment and following through to the end of its economic life, this manual explains how to efficiently maintain and operate different types of heavy equipment. Assigning heavy equipment to different projects and utilizing them in varied systems is a large part of construction operation; ensuring everything is monitored consistently and maintained accordingly is essential. This book aids engineers in facilitating straightforward, consistent reporting systems and cost-efficient equipment use. This is particularly of note to the construction industry.

Features:

- Enables engineers to save time and money on maintenance costs and maximize the availability of the heavy equipment
- Provides comprehensive coverage of methods and procedures for heavy equipment management
- Provides charts for practical use by engineers in the field, e.g., mapping out a maintenance schedule
- Includes chapters on maintenance and field operations organization, including safety and security organization

This book will be of interest to construction engineers, plant engineers, mechanical engineers, maintenance plant and field engineers.

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About the Author

From 1984 to 2013, **Ernesto A. Guzman** commuted to Jakarta, Indonesia, and Hail and Jeddah, Saudi Arabia, from Manila, Philippines, for much of his professional and academic concerns. Being a mechanical engineer, he worked as a heavy equipment expert for twelve years in the Philippines and then continued his path as an expat in Indonesia and Saudi Arabia for another twelve years on. He has been involved with construction projects at Philippine National Construction Corporation and general manager of Tierra Factor Corporation, both in the Philippines; Assistant Director for PT Bukaka Teknik Utama (Indonesia) for the Java steel tower transmission lines project; and general manager for PT Surya Mahkota Timber (Indonesia) for furniture manufacturing and project manager of Pt Eurasia Wood Industries (Indonesia) MDF-wood furniture manufacturing operations in Jambi, Sumatra, and Jakarta, Indonesia, among others.

In addition, he was also consultant to PT Sulmil Wood Industry (Sulawesi, Indonesia), Australia Consolidated Industries Insulation (Phil). And at Al Khodari and Sons, North Rail Project at Al Jouf, Saudi Arabia. He also taught at the University of Santo Tomas Graduate School (Philippines) lecturing on Production Management and Supply Chain Management. Then, he proceeded to lecture at the Royal Melbourne Institute of Technology (RMIT) International University in Ho Chi Ming City, Vietnam, also handling Supply Chain Management and Business Statistics.

He has written many papers, articles and manuals in various subjects from operations and maintenance of heavy construction equipment, quality and test procedure to methods of construction and quality assurance, material inspection and testing, including the article Industrialization for Development about the industrialization path for economic development of a nation, and on the light side, he has written the Ying and Yang in Contemporary Times.

He earned his Bachelor of Science in Mechanical Engineering at the University of Santo Tomas (UST) and his Master in Management degree at the Asian Institute of Management (AIM), both in the Philippines. His thesis was Corporate Strategic for Philippine National Construction Corporation (PNCC) Tollway Division. A study for the expansion of two main expressways, the North Luzon Expressway (NLEX), 83 km length, and the South Luzon Expressway (SLEX), 43 km length. He completed his academics for his PhD in Technology Management at the Technology University of the Philippine in Manila.

Introduction

Heavy equipment management is a major component of the construction industry. It can account alone for 57 percent of the total assent of a company. An equipment-proficient management system is paramount in order to be effective and profit-driven as a principal item of the corporation in the construction industry. Construction is a prime mover industry. It has been known to help the economy of a country sustain its economic growth during the period project work is in progress and after the project has been completed. In the construction business, the best equipment-managed company can be assured to have the most out of their heavy equipment.

An equipment management manual is one of the most tried and tested reference to attain this. Cost in equipment operations are most likely to end up with hidden and arbitrary cost left unidentified due to lack of structure and procedures in its operations. This can devour profit without the company board knowing what has happened. The best way is to have standard equipment procedures to minimize and eliminate wasteful operation practices.

The manual is a collection of heavy equipment working procedures that cover the equipment department organization, administration, maintenance and field operations of the equipment. From the period the equipment was purchased, assigned it first field project work and moved project to project up to the time the equipment shall have reached its economic life. The manual was prepared for ownership of at least 10–100 or more equipment in the fleet of a construction company.

It has been written to ensure the equipment technical and field personnel will clearly understand key procedures, information and technical concepts from the base level so as to apply and execute the procedures with ease and accuracy, leaving no room for error and misunderstanding, especially in the construction operations field where no indecision is tolerated.

Every chapter has a corresponding chart or form to accomplish the proper procedure outlined in the chapter. Each chart gives direct, factual and concise data to ensure the procedure is executed with accuracy. Each form will contain the exact data for each specific operation so as to provide the exact information for the engineer and manager on hand as well as for the equipment management officials in their command offices.

The number of these vital documents for equipment operation and maintenance (Figure 0.1):

- Fifteen (15) Charts and Tables.
- Twenty-Seven (27) Forms from equipment coding, manpower classification, branding, inspection, operations, PM schedules, time and cost, rental rate, safety and security coverage.

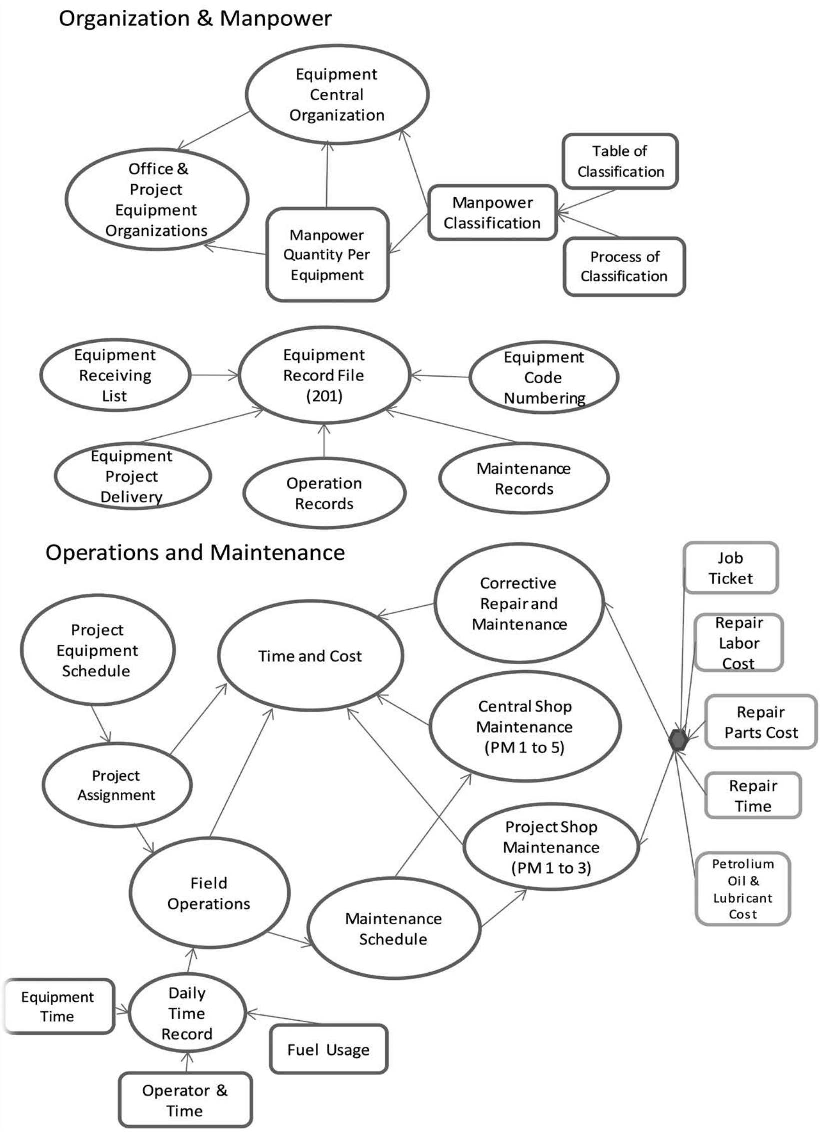

FIGURE 0.1 Equipment operation flow chart.

This book is the product of 20 years of work experience in the construction field as equipment manager to senior manager of construction companies that have completed projects such as hundreds of kilometers of expressways and roads, bridges, sugar mills, fertilizer manufacturing plants among other construction projects in the Philippines, Saudi Arabia, Indonesia and Malaysia.

Part I

Equipment Organization

1 Equipment Management Organization and Administration

1.1 GENERAL ORGANIZATION AND GENERAL FLOW CHART

One of the first important stages in equipment management is to have a well-organized organizational structure. There are two main structures, the centralized general organization where the central shop and yard and main equipment office are located and the satellite equipment project organization.

These tables of organization focus on policies and systems instead of people. The personnel assigned for each position have a distinctive function to avert any duplication. There is also a strong command responsibility of each from the manager to superintendents, to supervisor to engineers and down to the workforce of operators and maintenance crew.

Teamwork, collaboration and communication make the organization active and busy, when the projects are in operation. Equipment team members can rely on each other for guidance and support. This enables them to focus on the equipment operations' goals and carry out their duties and responsibilities effectively.

With a well-organized table of organization, the company can have as many as 10–15 different projects operating at the same time and located in different locations. The well-defined functions and positions will organize the equipment project operation efficiently and effectively.

Equipment Management Department shall be the sole responsible for all the company equipment, and this responsibility shall include administration, operations and maintenance functions of the equipment.

More specifically, these are the management and technical functions for purchasing, shipping, utilization, maintenance, repair, security, safety and disposal of equipment. Also included are the spare and repair parts sourcing, inventory and warehousing, central yard maintenance, central shop facilities and technical services.

To facilitate these functions, there are eight basic sections operating under the department. These are the central shop and yard, the technical services, field operations, personnel and administration, finance, accounting, purchasing and warehouse and inventory control. If it so warrant, another section can be separated, the transport and shipping section.

For some operations, a process plant can also be attached to the department. A department can be centralized or each project equipment operation can be created, according to the convenience of the projects locations an duration.

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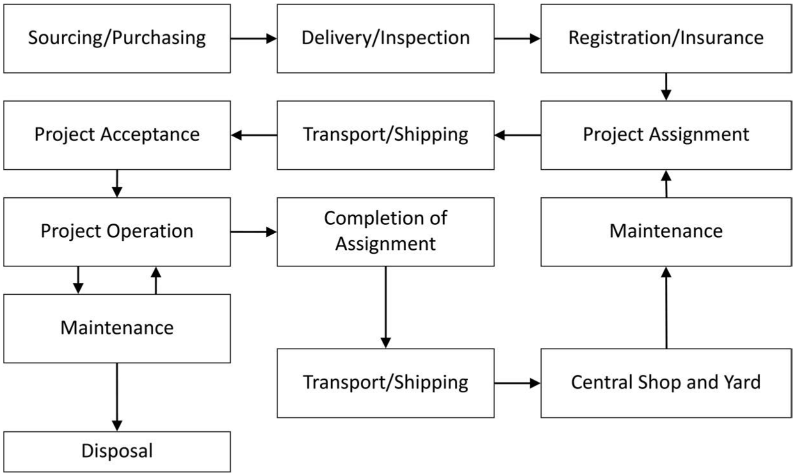

FIGURE 1.1 Equipment ownership flow chart.

A project can also be a centralized and a standalone if the project is large enough and the construction period long enough where equipment will be fully depreciated at the end of the project. Whichever way, a well-organized set is mandatory.

The flow chart of a simplified equipment ownership and operation of a construction company is shown in Figure 1.1.

2 Equipment Tables of Organization

2.1 CENTRALIZED ORGANIZATION

A company with a centralized equipment operation shall have all the functions of an independent organization. This shall act as the head office of equipment operation. It shall be composed of two major functions, the main functions which are the central shop and yard, the technical services and the field operations and the support functions which are the purchasing, warehouse, inventory control, personnel and transport.

If the equipment operation is a profit center, the finance with the cashier and the accounting shall be part of the equipment division. If the equipment operation is a cost center, these two functions will be located at the construction central office, but bookkeeping and a cashier will be part of the equipment central office.

Another option for a cost center operation is to locate the purchasing function with the central or head office of the company. But the warehouse and inventory control will be located with the central shop and yard location.

Whether it is a cost center or profit center, when a fleet equipment is deployed in a far out and separate project location, a project equipment organization shall be set up. This shall be called the equipment project section. A small set of technical people shall compose this project equipment field office. They shall be reporting both administratively to the equipment head office and operation wise to the project office they belong.

This is the reason documentation and very comprehensible organization functions be set up. While it may look simple, during actual project operation period, confusion can result if an unsystematic organization is set up adding to high hidden cost to the company (Figure 2.1).

2.2 PROJECT EQUIPMENT ORGANIZATION

The project equipment organization is a satellite unit of the equipment department located at the job site. It is designed to do field equipment operations and maintenance. It shall have a small warehouse and petroleum, oil, and lubricants (POL) distributor depot. A time and cost unit will carry out the equipment monitoring, and it shall submit all equipment time and cost reports to the technical services at the central office every month.

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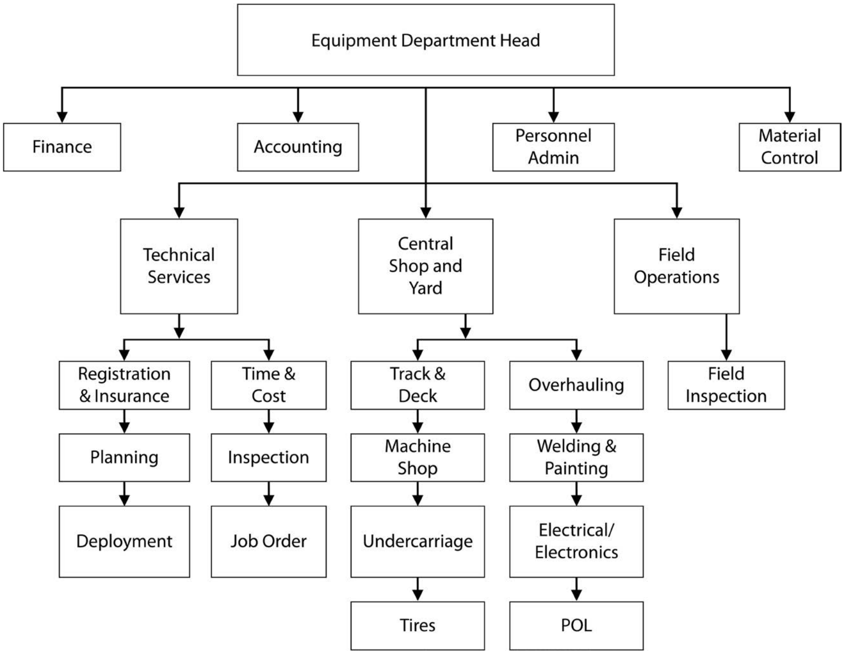

FIGURE 2.1 Centralized general table of organization.

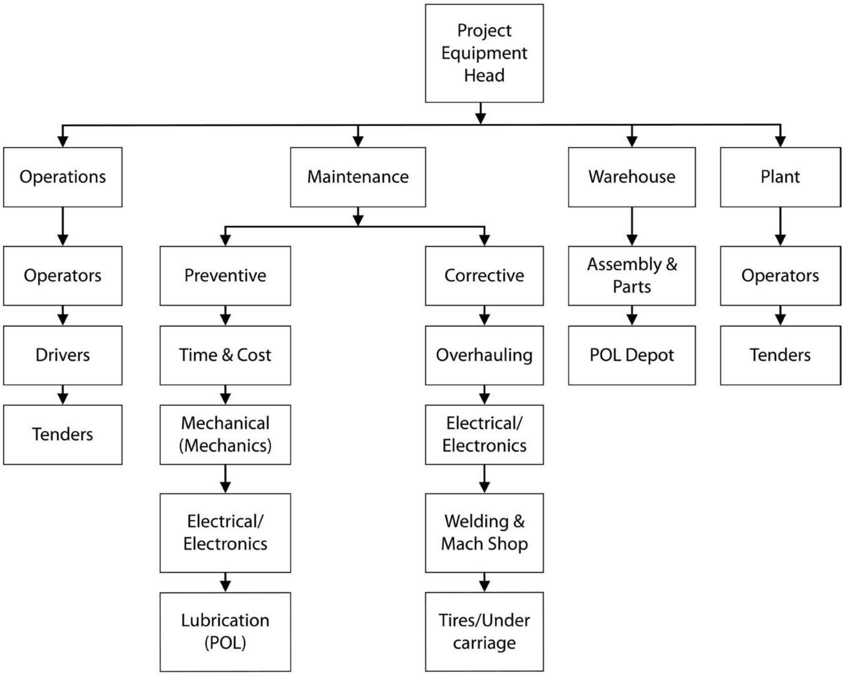

FIGURE 2.2 Typical project equipment table of organization.

This in turn shall be used to consolidate all the equipment utilization and availability reports for the management. The number of personnel in each set up shall depend on the number of equipment deployed in the project. This is shown in Section 3.4 in Chapter 3. The project equipment office is headed by a representative of the equipment head office. He is usually an Equipment Superintendent (Figure 2.2).